21

Lecture 21

Network Slicing

برش‌بندی شبکه

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 2 (Erlang B), 16, 19 and 20; normal distribution

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-21-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Define** a network slice and **decode** the S-NSSAI and the NSSAI sets used in signalling.
- **LO2** **Trace** slice selection at registration and slice-specific PDU session set-up.
- **LO3** **Explain** how slices are realised in the RAN, transport and core, and the isolation options.
- **LO4** **Describe** slice management: lifecycle, templates and the roles of operator and tenant.
- **LO5** **Quantify** the statistical-multiplexing cost of isolation, RAN resource partitions and slice admission control.

2 Prerequisite knowledge

You need the 5GC architecture and procedures (Lectures 19–20), the virtualisation concepts of Lecture 16, and the Erlang-B formula of Lecture 2. The multiplexing model uses the normal distribution: for a Gaussian load with mean μ and standard deviation σ, the capacity exceeded with probability ϵ is $\mu + z_{1-\epsilon}\sigma$ ($z_{0.99} = 2.326$).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Network slice	برش شبکه	Logical network providing specific capabilities and characteristics.
S-NSSAI	شناسهٔ برش	Single network slice selection assistance information (SST + SD).
SST / SD	نوع برش / متمایزکننده	Slice/service type and optional slice differentiator.
NSSAI	مجموعهٔ S-NSSAIها	Configured, requested, allowed and rejected sets.
NSSF	تابع انتخاب برش	NF that selects allowed slices and the AMF set.
NSSAA	احراز هویت ویژهٔ برش	Slice-specific authentication by a third-party AAA server.
NSAC	کنترل پذیرش برش	Limits on UEs or PDU sessions per slice.
Isolation	جداسازی	Degree to which one slice is protected from another.
Tenant	مستأجر	Customer (e.g. enterprise) for whom a slice is operated.
Slice template	الگوی برش	Standardised description of slice attributes (e.g. latency, throughput).

3 Introduction and motivation

The three IMT-2020 scenarios (Lecture 14) make conflicting demands: an eMBB service wants throughput and tolerates delay, a URLLC service wants guaranteed latency and reliability, an mMTC service wants huge numbers of cheap, rarely active connections. Building three separate networks would be uneconomic; running all on one network with best-effort sharing would fail the most demanding customers. *Network slicing* runs several logical networks on one physical infrastructure, each with its own capabilities, policies and, where needed, its own network functions [5], [6].

Slicing is also a business concept: an operator can offer a slice to an enterprise, a public-safety agency or a vertical industry as a product with guaranteed characteristics, managed through standard interfaces [1]. It is made possible by the technologies of Part III: virtualisation creates dedicated instances cheaply, SDN programs the transport, and the SBA lets NFs be selected per slice.

4 Fundamental concepts

4.1 Identifying slices

A slice is identified by an *S-NSSAI* (Figure 1): an 8-bit *slice/service type* (SST) and an optional 24-bit *slice differentiator* (SD) that distinguishes several slices of the same type (e.g. two enterprises' URLLC slices) [2]. Standardised SST values include 1 (eMBB), 2 (URLLC), 3 (massive IoT) and 4 (V2X), with further values added in later releases; operators may also define their own. A UE handles sets of S-NSSAIs:

- **configured NSSAI**: provisioned in the UE per PLMN;
- **requested NSSAI**: sent in the Registration Request;
- **allowed NSSAI**: returned by the network for the registration area (at most 8 S-NSSAIs);
- **rejected NSSAI**: requested slices refused, with a cause (e.g. not available in this area).

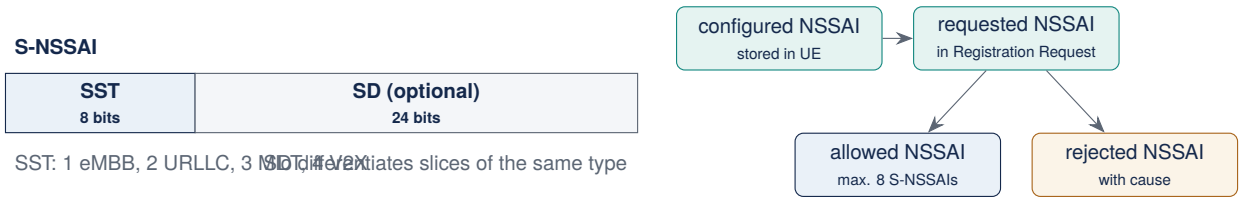


Figure 1. S-NSSAI structure and the NSSAI sets exchanged at registration.

4.2 What a slice consists of

An end-to-end slice spans the RAN, the transport network and the core (Figure 2). Some functions are typically shared (AMF, NSSF, NRF, UDM, the RAN nodes), others may be slice-specific (SMF, UPF, PCF, application functions). A UE can use up to eight slices simultaneously, but all through the same AMF; each PDU session belongs to exactly one slice.

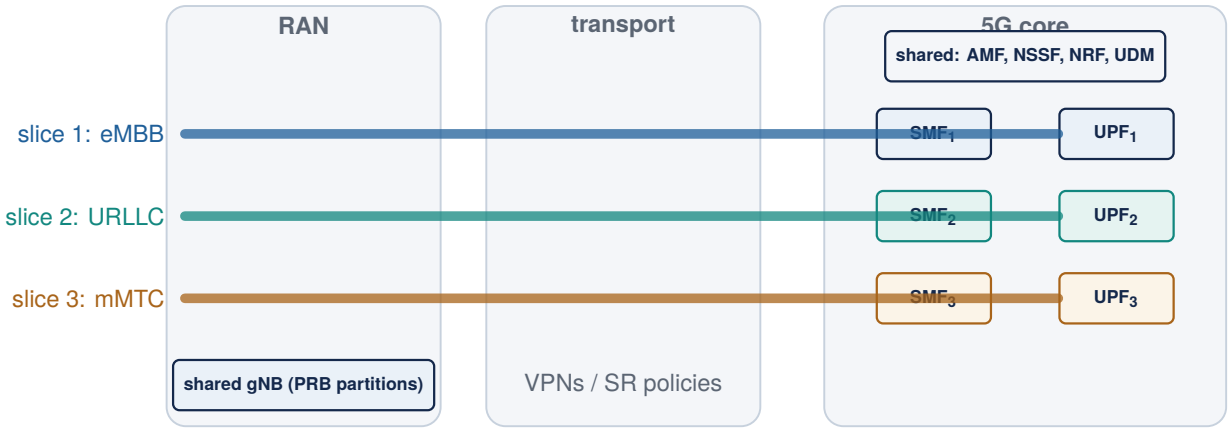


Figure 2. Three end-to-end slices sharing RAN and common control functions, with slice-specific SMFs and UPFs.

4.3 Isolation

Isolation can be achieved at several levels: *logical* (separate policies and QoS on shared NFs), *virtual* (dedicated NF instances on shared infrastructure) or *physical* (dedicated hardware, spectrum or sites). Stronger isolation protects a slice from others' load, faults and security incidents, but forfeits the statistical-multiplexing gain of sharing (Section 6.1). The right level depends on the service-level agreement: a public-safety slice may need dedicated UPFs and reserved radio resources, while a consumer video slice can share almost everything.

5 Detailed technical explanation

5.1 Slice selection

At registration the AMF checks the requested NSSAI against the subscribed S-NSSAIs from the UDM and, if necessary, asks the NSSF, which returns the allowed NSSAI for the UE's location and possibly a different AMF set better suited to the slices (Figure 3). The AMF may then re-route the registration to an AMF of that set. When the UE requests a PDU session for an S-NSSAI and DNN, the AMF discovers an SMF serving that slice through the NRF, and the SMF selects a UPF of the slice. Slice information is also given to the RAN, which may use it for resource allocation and mobility decisions.

5.2 Slice-specific authentication and admission control

Release 16 added *network slice-specific authentication and authorisation* (NSSAA): after primary authentication, access to a slice can additionally depend on credentials verified by an AAA server of the slice tenant (e.g. an enterprise), so that the tenant controls which devices use its slice. Release 17 added *network slice admission control* (NSAC): the network can limit the number of registered UEs and of PDU sessions per slice, enforcing the capacity sold to a tenant [5].

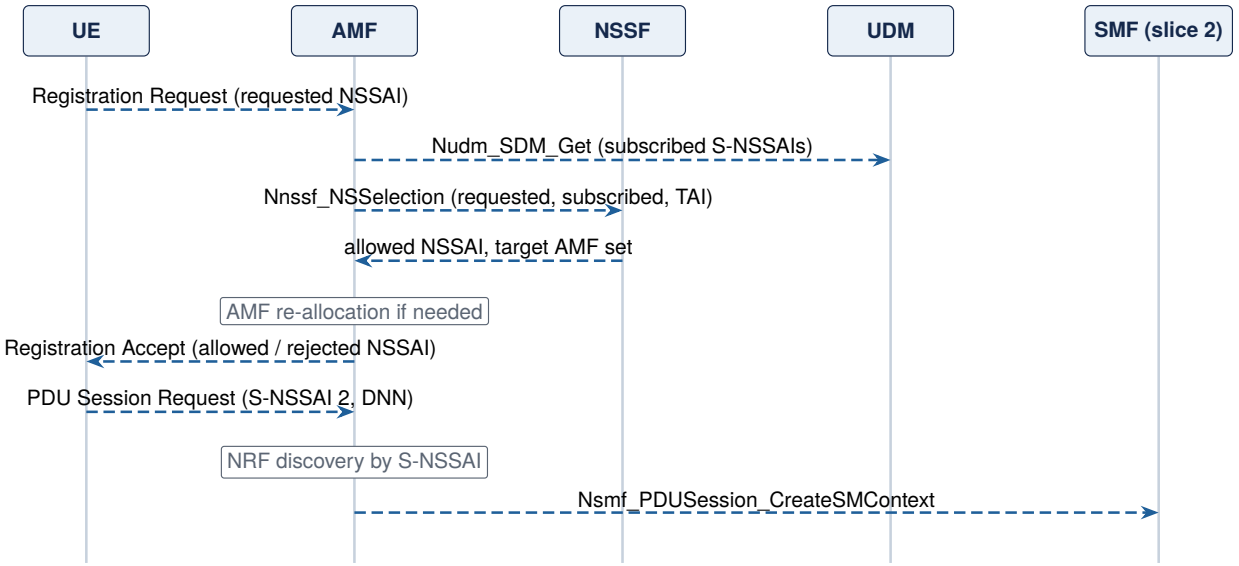


Figure 3. Slice selection at registration and slice-specific PDU session set-up (simplified).

5.3 Slicing in the RAN and transport

The RAN is shared; slices are realised by scheduling policies: minimum, maximum and dedicated shares of physical resource blocks per slice, slice-aware admission control and priority, and possibly separate bandwidth parts or carriers. The transport network maps slices to virtual private networks, segment-routing policies or hard-isolated channels with guaranteed bandwidth and latency. Without these, a core slice alone guarantees nothing end to end.

5.4 Slice management

3GPP management specifications describe slices as managed objects: a *network slice* composed of *network slice subnets* (RAN, transport, core), with a lifecycle of preparation, commissioning, operation and decommissioning [1]. Customers' requirements are expressed with slice templates – for example the GSMA's generic network slice template lists attributes such as throughput, latency, availability, number of devices and isolation – which the operator's orchestration translates into NF instances, policies and RAN and transport configuration, using NFV MANO (Lecture 16) and SDN controllers (Lecture 15).

6 Mathematical models

6.1 The cost of isolation

Model. Slice i has a Gaussian load with mean μ_i and standard deviation σ_i, independent of the others. To keep the probability of overload below ϵ, dedicated capacity per slice and shared capacity for all slices are

$$C_i = \mu_i + z\,\sigma_i, \qquad C_{\text{shared}} = \sum_i \mu_i + z\sqrt{\sum_i \sigma_i^2}, \tag{1}$$

with $z = z_{1-\epsilon}$. The *multiplexing gain* of sharing is

$$G = \frac{\sum_i C_i}{C_{\text{shared}}} \geq 1. \tag{2}$$

Example. Three slices with $\mu = 100$, $\sigma = 30$ (in any capacity unit), $\epsilon = 1\,\%$ ($z = 2.326$): dedicated $3 \times 169.8 = 509.3$, shared $300 + 2.326\sqrt{3} \times 30 = 420.9$; $G = 1.21$ – isolation costs 21 % more capacity (Figure 4).

Interpretation. The gain grows with the number of slices and with the burstiness σ/μ; it vanishes for constant loads. Isolation is therefore expensive exactly for bursty services – the economic argument for sharing what can be shared and isolating only what the SLA requires. Correlated loads (e.g. all slices peaking in the same hour) reduce the gain.

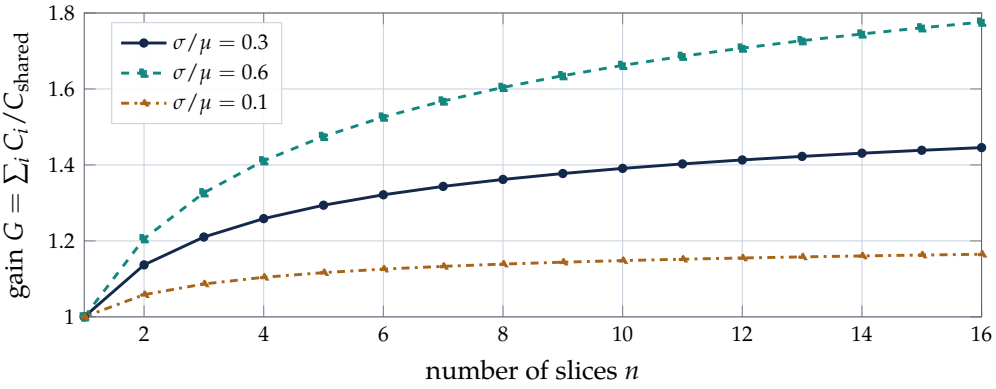


Figure 4. Multiplexing gain of a shared pool versus dedicated slice capacity (Eq. (2), $\epsilon = 1\,\%$).

6.2 RAN resource partitions

Model. A cell has N resource blocks; slice i is guaranteed a minimum share ρ_i and, with a robust modulation and coding scheme, a rate r_i per resource block. Its guaranteed throughput is

$$R_i^{\min} = \lfloor \rho_i N \rfloor \, r_i, \tag{3}$$

while unused guaranteed resources may be lent to other slices unless they are *dedicated*.

Example. $N = 273$ (100 MHz at 30 kHz), URLLC minimum 20 % (54 resource blocks), $r = 0.5$ Mbit/s per block: $R^{\min} = 27$ Mbit/s, sufficient for many control flows, with the rest of the cell available to eMBB when URLLC is idle.

6.3 Slice admission control

Model. If a tenant's slice is limited to c simultaneous PDU sessions and sessions arrive as a Poisson stream with offered load A Erlang, the probability that a new session is rejected is the Erlang-B formula of Lecture 2:

$$B(A, c) = \frac{A^c / c!}{\sum_{k=0}^{c} A^k / k!}. \tag{4}$$

Example. $A = 40$ E: with $c = 50$, $B = 1.9\,\%$; with $c = 60$, $B = 0.07\,\%$. A quota of $c = 53$ is the smallest giving $B \leq 1\,\%$. The quota is part of the contract: it protects other slices and defines what the tenant pays for.

7 Network architecture and diagrams

Figure 2 shows a common deployment pattern: shared RAN with per-slice scheduling policies, transport VPNs per slice, shared AMF and subscriber data, and per-slice SMF and UPF – the URLLC slice's UPF placed at the edge, the eMBB slice's UPFs centrally. An enterprise slice may additionally have a UPF on the enterprise premises (Lecture 19) and NSSAA towards the enterprise's AAA server. The management system keeps the mapping from each slice to its subnets and resources.

8 Step-by-step procedures

Procedure 1 • Registration with slice selection

1. The UE sends a Registration Request with the requested NSSAI (from its configured NSSAI).
2. The AMF retrieves the subscribed S-NSSAIs from the UDM.
3. If it cannot decide alone, the AMF queries the NSSF with requested and subscribed NSSAI and the UE's TAI; the NSSF returns the allowed NSSAI and possibly a target AMF set.
4. If another AMF set is needed, the registration is re-routed; slices requiring NSSAA are authorised with the tenant's AAA server.
5. Registration Accept returns allowed and rejected NSSAI; the UE uses only allowed S-NSSAIs in PDU session requests.

Procedure 2 • Commissioning an enterprise slice

1. The tenant's requirements are captured in a slice template (throughput, latency, devices, area, isolation, availability).
2. The operator's management system decomposes the slice into RAN, transport and core subnets and checks feasibility.
3. NFV orchestration instantiates slice-specific SMF/UPF (e.g. an on-premises UPF); SDN controllers set up transport paths; RAN policies reserve resource shares in the covered cells.
4. Subscriptions of the tenant's devices are updated with the new S-NSSAI; the NSSF configuration and NSAC quotas are updated.
5. The slice is activated and its performance is monitored against the SLA.

9 Worked examples

Worked example 1 • Decoding S-NSSAIs

Problem. A UE's allowed NSSAI contains (SST 1), (SST 2, SD `0x00A1B2`) and (SST 2, SD `0x00C3D4`). Interpret it.
Solution. One default eMBB slice without differentiator and two URLLC slices distinguished by their SDs – for example two different enterprise or industrial customers. The UE can open PDU sessions in all three slices simultaneously through the same AMF.

Worked example 2 • Isolation cost

Problem. Using Eq. (1), how much capacity does an operator save by sharing UPF capacity among the three slices of Section 6.1 instead of dedicating it, and what does the URLLC tenant risk?
Solution. $509.3 - 420.9 = 88.4$ units (17 % of the dedicated total). The URLLC slice then competes with the others during joint peaks; if its SLA requires guaranteed performance independent of others' traffic, its share must be reserved (logical isolation with priority) or dedicated (virtual isolation).

Worked example 3 • Dimensioning a quota

Problem. A logistics company expects 40 E of simultaneous sessions in its slice and accepts 1 % blocking. What NSAC quota should be configured and how does the blocking change if its load grows by 12.5 %?
Solution. $c = 53$ (Section 6.3). At $A = 45$ E with $c = 50$ the blocking would be 5.4 %; with $c = 53$ it is 3.1 % – still well above 1 % – the quota (and price) must be renegotiated as the tenant grows.

10 Technology comparison

Table 1. Ways of differentiating service in mobile networks.

	QoS classes (4G/5G)	Dedicated core (DECOR, 4G)	Network slicing (5G)
Granularity	Per flow/bearer	Per UE (core selection)	Per PDU session, end to end
Scope	Packet treatment	Core NFs	RAN, transport, core, management
Isolation	None (priorities)	Core instances	Logical to physical
Per-UE use	Many flows	One dedicated core	Up to 8 slices
Customer view	Not exposed	Not exposed	Product with SLA, tenant management

Slicing and QoS. Slices and QoS flows (Lecture 22) are complementary: a slice defines *which* network and resources serve a session; QoS flows define how packets *within* it are treated. A

URLLC slice still uses QoS flows with appropriate 5QIs.

11 Key takeaways

Key takeaways

- A slice is an end-to-end logical network identified by S-NSSAI = SST (8 bits) + optional SD (24 bits).
- NSSAI sets: configured, requested, allowed (max. 8), rejected; NSSF selects slices and AMF sets.
- Slices combine shared and dedicated NFs; RAN and transport must also be sliced for end-to-end guarantees.
- Isolation costs multiplexing gain: $G = \sum(\mu_i + z\sigma_i) / (\sum \mu_i + z\sqrt{\sum \sigma_i^2})$.
- RAN partitions give $R_i^{\min} = \lfloor \rho_i N \rfloor r_i$; NSAC quotas are dimensioned with Erlang B.
- NSSAA and NSAC let tenants control access and operators enforce contracted capacity.

12 Exercises

Exercise 1 • Level 1 – conceptual • Slices or QoS?

For each requirement, say whether a separate slice or a QoS flow within an existing slice is the better tool: (a) prioritising video calls over downloads for consumers; (b) a police network that must keep working when the public network is congested; (c) a factory whose data must not leave its premises.

Exercise 2 • Level 1 – conceptual • Isolation levels

Give an example of logical, virtual and physical isolation in the core and in the RAN.

Exercise 3 • Level 2 – numerical • Bursty slices

Repeat the calculation of Section 6.1 with $\sigma = 60$. What is the gain, and what does it mean for slices carrying bursty traffic?

Exercise 4 • Level 2 – numerical • RAN share

A URLLC slice is guaranteed 15 % of 273 resource blocks with a robust MCS of 0.2 Mbit/s per block. Compute its guaranteed rate. How many 64 kbit/s control streams fit, and what happens to unused blocks if the share is “minimum” rather than “dedicated”?

Exercise 5 • Level 3 – analytical • Quota design

Show that the smallest quota with $B \leq 1\,\%$ at $A = 40\,\mathrm{E}$ is $c = 53$ (use the recursion $B(A,k) = AB(A,k-1)/(k+AB(A,k-1))$). Then estimate the quota needed for $A = 80\,\mathrm{E}$ and compare the ratio c/A; explain the trend.

Exercise 6 • Level 3 – analytical • Designing a slice

A hospital wants a slice for 200 connected devices (monitors, imaging) with 10 ms latency, 99.99 % availability inside its campus, and data that must stay on site. Propose the S-NSSAI, the shared and dedicated components, the UPF placement, RAN measures and admission/authorisation features.

Solutions

Solution to Exercise 1

(a) QoS flows (different 5QIs) in the consumer slice. (b) A slice with reserved RAN resources, priority and preemption (and possibly dedicated core NFs) – isolation from public load is the requirement. (c) A slice with an on-premises UPF and local breakout (possibly NSSAA), since the requirement is about where the user plane terminates, not packet priority.

Solution to Exercise 2

Core: logical – shared SMF/UPF with per-slice policies; virtual – dedicated SMF/UPF instances on shared servers; physical – dedicated servers or site. RAN: logical – per-slice priority in a shared scheduler; virtual – reserved/dedicated PRB shares or a separate bandwidth part; physical – a dedicated carrier or dedicated small cells.

Solution to Exercise 3

Dedicated $3 \times (100 + 2.326 \times 60) = 718.7$; shared $300 + 2.326\sqrt{3} \times 60 = 541.7$; $G = 1.33$. The burstier the traffic, the more capacity isolation wastes: bursty best-effort slices should share; isolation should be reserved for slices whose SLA demands it.

Solution to Exercise 4

$\lfloor 0.15 \times 273 \rfloor = 40$ blocks (41 if rounded), $R^{\min} = 40 \times 0.2 = 8\,\mathrm{Mbit/s}$: 125 streams of 64 kbit/s (fewer with headroom). With a *minimum* share, unused blocks are lent to other slices and reclaimed when URLLC traffic appears (possible short scheduling delay); with a *dedicated* share they stay idle, giving the strongest guarantee at a capacity cost.

Solution to Exercise 5

The recursion gives $B(40,52) = 1.1\,\%$ and $B(40,53) = 0.82\,\%$, hence $c = 53$ ($c/A = 1.33$). For $A = 80\,\mathrm{E}$ the recursion gives $c = 96$ ($c/A = 1.2$): larger pools need relatively less headroom – the same economy of scale as in trunk dimensioning (Lecture 2) and the multiplexing gain of Section 6.1.

Solution to Exercise

S-NSSAI: SST 2 (URLLC) or an operator-specific SST, with an SD identifying the hospital. Shared: AMF, NSSF, UDM. Dedicated: SMF and a UPF on the hospital campus (local breakout, data on site), possibly redundant UPFs for 99.99 %. RAN: indoor small cells with a minimum or dedicated PRB share and slice priority; latency via 30 kHz numerology or mini-slots. Access: NSSAA with the hospital's AAA server; NSAC quota of about 200 UEs; transport path with guaranteed bandwidth. Monitoring of latency and availability against the SLA.

13 Further reading

Slicing in the 5G system is specified in TS 23.501 [5] (clause 5.15) with procedures in TS 23.502 [3]; S-NSSAI encoding in TS 23.003 [2]; management concepts in TS 28.530 [1]; service requirements in TS 22.261 [4]. Foukas et al. [6] survey slicing architectures and research questions; Rommer et al. [7] cover slicing in the 5G core.

References

[1] 3GPP, *Management and orchestration; concepts, use cases and requirements*, TS 28.530, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/28530.htm.

[2] 3GPP, *Numbering, addressing and identification*, TS 23.003, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23003.htm.

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[7] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.